

Question Number	Answer	Mark
8(a)(i)	The only correct answer is B endemic <i>A is not correct because the correct term is endemic</i> <i>C is not correct because the correct term is endemic</i> <i>D is not correct because the correct term is endemic</i>	(1)

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	An explanation that makes reference to the following points: • each population of island fox cannot mate with individuals on another island (1) • (as each island has) different {(environmental) conditions / habitats / selection pressures} (1) • individual with advantageous allele will (survive and) {reproduce/pass onto offspring} (1) • (resulting in) populations being genetically different (to each other) / offspring on one island would be genetically different to offspring on a different island / accumulation of different genetic material results in new subspecies (1)	accept there is {geographical / reproductive} isolation / allopatric speciation accept they can't reproduce together accept foxes can only mate with foxes on their own island / no gene flow between islands ignore can't reproduce to form fertile offspring accept named conditions accept natural selection occurs / change in allele frequency accept individuals without advantageous alleles will not survive accept inbreeding (depression) occurs / decrease in gene pool ignore gene accept different {alleles / mutations} would be advantageous (on different islands) accept {different alleles will be {selected for / lost} / different change in allele frequency} (on different islands) accept (populations on different islands have) different gene pools ignore gene / speciation ignore change in allele frequency	(4)

Question Number	Answer	Additional guidance	Mark
8(a)(iii)	An explanation that makes reference to the following points: {comparison / analysis} of (the sequences in) named biological molecules / molecular phylogeny (1) (therefore) the subspecies with the {most similarities / fewest differences} (are the most closely related to the Santa Cruz island fox) (1)	e.g. DNA, mRNA, proteins, alleles, base sequence / introns ignore bases unqualified accept comparison of banding pattern in gel electrophoresis / DNA profiling ignore more similar means more related	(2)

Question Number	Answer	Additional guidance	Mark
8(b)(i)	A calculation in which: correct percentage change (1)	<u>Example of calculation</u> $1500 - 80 = 1420$ $(1420 \div 1500) \times 100 = (-)94.67 / 94.7 / 95\%$ accept 1994 values between 1500 to 1520 inclusive to give answer between 94.67 to 94.74 inclusive or rounded to 95(%) max 2 dp correct answer scores full marks	(1)

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	An explanation that makes reference to the following points: - calculate the allele frequency for {p / q} (in 1994) / calculate $p^2 + 2pq + q^2 = 1$ (in 2000) (1) {compare this with / subtract} the {(allele) frequency / p / q} (in 1994) (to see if there has been a change) (1)	accept calculate the {dominant / recessive} allele frequency ignore calculate allele frequency unqualified (i.e. not linked to dominant/recessive/p/q/correct equation) accept calculate $p+q=1$ reject mp1 if an equation is incorrect	(2)

Question Number	Answer	Additional guidance	Mark
8(b)(iii)	An answer that makes reference to four of the following points: - establish protected areas / increase food source (1) - reduce the population of {golden eagles / dogs / racoons} (1) (reduce deaths from disease by) {treating diseased foxes (with medication) / vaccination} (1) - use of captive breeding programme (to increase population size) / reintroduce (captive bred) foxes (1) {analysis of genotype of foxes used in captive breeding programme / use of stud books} to ensure genetic diversity of fox population is maintained (1)	accept provide food / protect from predators (including humans) education of local population to protect {fox / fox habitat} ignore {establish / put in a} zoo ignore remove predators unqualified accept {eradicate / cure} disease ignore prevent disease transmission accept breeding programme accept description of captive breeding programme accept artificial insemination / exchange of animals with different zoos ignore breeding with similar foxes / other fox populations (from different islands) accept select foxes with different alleles to mate together / prevent inbreeding	(4)